Imperial College London

AERO96005 - GROUP DESIGN PROJECT

Wholly Reusable Launch System (Recovery)

Flight Simulation (Performance) Of A Novel Synchrocopter Concept For The Aerial-Recovery Of Falling Rocket Stages

SYNCHRO CAPTURE

Student: Tze Leung WONG
CID: 01339554

Project Supervisor: Dr. E. Levis
Team Supervisor: Dr. E. Levis

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Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ

Abstract

To simulate the performance of the new design, X-Plane 11 Flight Simulator was used to model the helicopter, and subsequently the prototype were tested. The target was to validate the theoretical calculation from different teams, and to provide feedback at the end of the project. To fully recreate the geometry of the lifting surface such as the blades was proven to be a challenge, and therefore modification has been adopted to provide similar aerodynamics characteristics.

This report focuses on the performance of the helicopter. Four tests were carried out to ensure that the flight model can meet the requirement of the mission profile. Vertical and Forward speed test have showed an adequate agreement with the theoretical solution with acceptable error. However, there is disagreement for the autorotation and hovering test mainly due to limitation and modelling process of X-Plane. Overall, the helicopter meets the requirement of the mission, however more tests should be conducted as future work to fully understand the behaviour of the model.

Figure 1: CAD Model of final design

Acknowledgements

I would like to express my sincere gratitude to my supervisors Dr. Errikos Levis and Dr. Maria J D Ribera Vicent for their invaluable guidance during this project. Rotorcraft dynamics is an unfamiliar topic for us, and such novel concepts are proven to be challenging. Undoubtedly this project would not be successful without their support. It is my pleasure to work with these experts in the helicopter and simulation field.

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra **Atri Sharma** Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz **Sam Jenner** Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston **Hakan Serpen** Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

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1 Introduction

Designing a successful Wholly Reuseable Launch System has been a challenge in aerospace industry, as it significantly reduces the cost of launching payloads into orbit. Therefore this year the project were to challenge current students to re-design this system[1]. As part of the recovery system design teams, it involved the design of a recovery vehicle to capture the re-entry vehicle from the launcher team. This technical report focuses on the flight simulation and testing stage (WRR06). It mainly presents the software selection process and flight test results. The main role of WRR06 involves generating a realistic flight model, modelling capture process and performing flight test. Ultimately this sub-team is responsible for verifying the analytic data from the other sub-teams, and to ensure that the aerial vehicle can satisfy the standard requirement from FAA regulation[2][3].

2 Software Overview

2.1 X-Plane vs FlightGear

At the early stage of the project, two candidates were shortlisted for the software selection. X-Plane (XP) and FlightGear (FG) was chosen due to its popularity and flexibility amongst the researchers at the Department of Aeronautics as well as Imperial College London. The aforementioned simulators has produced a fairly realistic simulation and are able to carry out flight test on new model. However, their mechanics behind the simulation engine involves the use of different approaches.Therefore, decisions must be made carefully to avoid irreversible and unnecessary workload for the team. This chapter will briefly explain the principle behind these simulators and the selection criteria for the most suitable software.

2.2 X-Plane

X-Plane uses Finite Element method ”Blade Element Theory” (BET) to calculate all the necessary data to predict the performance of the model[4]. Force vectors are generated in multiple elements on each lifting surface, and the calculation iterates at least 15 times per seconds. In addition, developers also created Airfoil-maker and Plane Maker as an editing tool for the XP along. User can then define the geometry of each surface by assigning aerofoil series at the tip and the root sections. XP will determine the aerodynamics characteristics of finite wing based on 2D characteristics curves and the empirical correction table. Its user-friendly interface allows inexperience users to design aircraft from scratch. The biggest advantage of XP is the user does not need to entail incompatibility between the simulation engine and the third party programs. In addition, there is less numerical data required to input and user can customize the geometry of the aircraft model easily. The limitations of this software will be discussed in later section.

2.3 FlightGear

FlightGear is an open source program, and emphasis on the highly customisable aircraft, which allows the public to develop new ideas for the new aerial vehicle. In contrast to XP, FG relies on its Flight Dynamics Model (FDM) to predict the flight mechanics based on its geometry (YASim) or stability derivatives (JSBSim)[5]. The user would require enough background knowledge and an extensive research before modeling in order to calculate all the stability derivatives. Then the FDM can predict the performance of the model based on these parameters. These FDM can be fairly accurate, if a wind tunnel or an actual flight data is known. This is due to the involvement of empirical correction factor, which is not ideal for the new model.

2.4 Comparison

A few major differences between XP and FG are listed in Table 2. In addition, the preference has been labelled in the last column based on the selection criteria set by WRR06 and to be discussed in the next section. The X-Plane usually verifies the data from other teams through a flight test, and the FlightGear is dependent on the stability derivatives from the other teams for the testing of aerial vehicle.

	XP	FG	Preference
Simulation Mechanics	BET	Geometry/Stability Derivative	XP
Experimental Data	No	Preferred (to optimize the model)	XP
Model Visualization	Yes	No	XP
Capture Mechanism (Plugin)	Less Flexible	Highly Flexible	FG
Integration with MATLAB	Yes	Yes	FG/XP

Table 2: Comparison between X-Plane and FlightGear

2.5 Final Decision

Having understood and distinguished the features between them, WRR06 has outlined the selection criteria of the simulation software for this project.

1. Accuracy
   - XP provides a more realistic data using real-time finite element analysis. However, FG is accurate only if there is a sufficient experimental data available.
2. Data Validation
   - Instead of relying on the data from other teams, WRR06 prefers to validate the data. This can simplify the communication between the teams significantly and less data required to be calculated . Increasing the complexity of the model will lead to an increase in manpower and time. It is therefore not feasible in this project due to the size of the team and the time-frame.
3. Historical Data and Successful Example
   - 2 FDMs (JSBSim and YASim)[6][7] in FG cannot be fully utilized due to lack of wind tunnel data and real flight data respectively. In addition, towards the end of the conceptual stage, an unmanned synchropter (helicopter with intermeshing rotors) has been chosen, in which there is a few successful model in either of the software (e.g. K-1200 K-MAX). WRR06 believes FG can increase the complexity of the model and it is not feasible due to the same reason stated in second criteria "Data Validation".
4. Capture Mechanism
   - Although the capture mechanism (plugin) is easier to implement in FG, its accuracy cannot be simulated if the aircraft is poorly modelled as stated in the second criteria "Data Validation".

Based on these 4 main considerations, WRR06 chose X-Plane as the most suitable software.

2.6 Limitation of X-Plane (v.11)

Even though X-Plane shows a highly realistic model, it is worth to highlight the limitation of this software and how it will impact the flight test. The discussion of these limitations will be revisited after the flight test has been carried out in later chapter.

1. Fuselage Effect
   X-Plane 11 does not consider the fuselage as part of the lifting body, but drag body due to the complex modelling process[8].

2. Intermeshing Blade effect
   X-Plane 11 cannot be accounted to the blade-to-blade interaction of the helicopter[9]. These blades were treated as a separate wing body, and the simulation can only output the resultant force vectors. Unfortunately, there is no alternative solutions or available plugins due to a limited time and available online resources, which this should also be considered.

3. Aerofoil Modelling
   During the aerofoil modelling process, Airfoil-maker only requires the user to input the critical data and fine tuning at the end[10]. Therefore, it cannot resemble the exact aerodynamics characteristics of the aerofoils. As a result of this, the aerofoil is only an estimation and cannot be accurately created. To obtain a better data, a few methods have been implemented which will be discussed in the next section.

4. Capture Process
   The main target of this mission is to capture the re-entry vehicle in the mid-air. This was proven to be the most challenging part of this project due to the complexity in the simulation software. Slung load method has been used and a external plugin is created in order to simulate the capturing process[11]. This method was agreed amongst the team as the quickest and the most realistic method within this time-frame and teammates' ability. An impulse force can act on the helicopter from a certain distance and can become a constant force at post-capture stage.

3 Modelling Process

Modelling is the most crucial part before the simulation experiment. Precise geometry and configuration must be made in order to resemble the most realistic performance of the model. Plane Maker and Aerofoil-maker are programs bundled with X-Plane, and it can be easily integrated together for the simulation. This allows the user to alternate the physical specifications of the aircraft, in particular the aerofoil geometry and the aircraft's body[10][12]. This chapter will discuss the modelling process. The final simulation model is shown in figure 2.

(a) Power Off

(b) Cruising

Figure 2: Finalized model for testing

3.1 Aerofoil

Although there is no complete aerofoil database for the X-Plane, airfoil-maker is an useful tool to customize and design a new aerofoil. By importing key 2D aerodynamics characteristics, the software can produce a good estimate of a 3D finite wing. However, it should be noted that this program does not produce an exact geometry and the characteristics are based on a pre-defined value from the user. It provides a good estimate of the aerofoils' behaviour if sufficient nodes are created. In this project, four different aerofoils are used in the propellers, horizontal and vertical stabilizers. These decision were made by the WRR02 team, as aerofoil series optimizes the aerodynamics performance. Before the data were obtained, the airfoil-maker has supplied two aerofoils for this model, namely NACA 0012 and NACA 23012. Therefore, only data for NACA 4418 and NACA 63206 are to be determined by the different methods. Initially, the idea of generating .afl file directly based on XFOIL[13] and JAVAFOIL[14] to save time and workload. However, due to its low accuracy at high AoA this has been abandoned[15], and experimental aerofoil data was used as the basis of aerofoil creation. The experimental points from TR824[16] have also been confirmed by the Computational Fluid Dynamics (CFD) Method from WRR02. Figure 3 illustrates the differences of the three sources for NACA 4418. This process was also repeated for NACA 23012. Table 3 shows the summary of all the aerofoil used in this project.

aerofoil Serie	Component	Data Sources
NACA 23012	Root of Blade	TR824
NACA 63206	Tip of Blade	TR824
NACA 4418	Horizontal Stabilizer	X-Plane (supplied)
NACA 0012	Vertical Stabilizers	X-Plane (supplied)

Table 3: Summary of the aerofoils

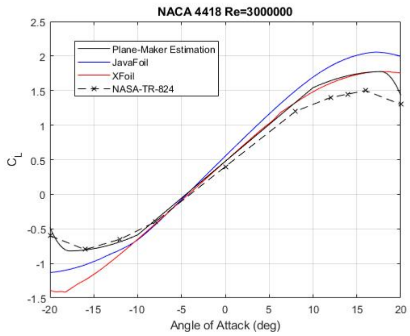


Figure 3: C_L curve from different sources

3.2 Propeller and Engine

Since X-Plane uses Blade Element Theory to predict the performance of the helicopter, it is important to model the propellers as accurately as possible. In this project, two propellers were generated by a single free turbine engine and rotates in opposite direction. WRR02 provides all numerical data such as the geometry and aerodynamics characteristics of the blades. This consists of two different aerofoils, one at each end and the transition occurs at 90% of the rotor radius. In addition, BERP rotor design is used at the tip section of the blade to reduce the tip Mach number and reduce the formation of drag force at the notch area. The geometry of the blade is shown in Figure 4. The twist angle is entered manually due to non-linear distribution and the drag-divergence Mach number M_{dd} is given as 0.85 as suggested by the WRR02[17]. However, due to the limited element available in plane-maker, the blade is modelled as a rectangular section with constant root aerofoil as shown in Figure 5. Firstly, the chord length cannot be customized in the "prop spec" section, and secondly it does not allow the user to set the location of aerofoil transition. Changing the tip aerofoil as NACA 63206 will force the X-Plane to start the transition in the middle of the blade. Since NACA 63206 has a lower lifting characteristic curve, the total lifting of the blade can significantly reduce the model. The principle behind the BERP tip design is complex. This is due to the combination of the notch, thin tip section and high sweep angles to optimise the performance [18]. Since the plane maker cannot produce a detailed geometry at 0.9r/R section, creating a sweep angle will only reduce the wave drag at the tip section. Wave drag only becomes significant in a high speed forward flight, where the advancing side experience a high Mach number[19]. Therefore, the sweep angle was not modeled as the helicopter will not fly at very high speed. To be noted, the lift and drag characteristics will not be affected at the lower speed flight.

Lastly, configuration of the turbine engine were provided by WRR04[20]. Parameters such as maximum available power and specific fuel consumption determines the performance of the helicopter during testing and will be compared with the prediction made by the WRR02 and WRR05 teams.

3.3 Control Surface

The empennage sizing is determined by the WRR02[17]. Two vertical stabilizers were located at the tip of the horizontal stabilizer to form a H-shape tail. The horizontal and vertical stabilizer were discretized into ten elements and control surfaces are equipped in certain elements. Rudders and elevator were attached to enhance the maneuverability of the helicopter in pitch and yaw direction at 0.3 of the chord ratio. These values were chosen based on the historical data from different helicopters.

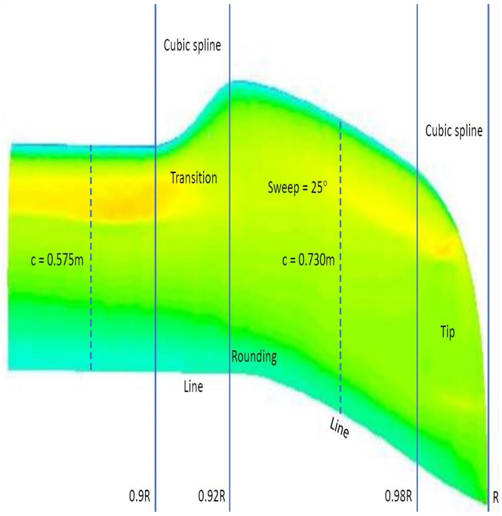


Figure 4: BERP Blade Design

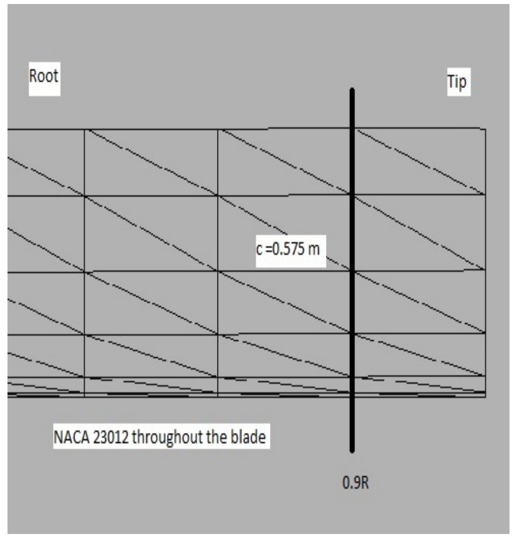


Figure 5: Finalized Blade Model

4 Flight Test

Once the helicopter was modelled and provided with sufficient data, the test can be carried out in a ground (IGE) and an out of ground effect (OGE). Due to nature of the mission, the test only focuses on OGE. Its location was assumed as one rotor diameter above the ground, and the wind speed is absolutely zero. Flight test in the low altitude (IGE) should be performed if time permits. Since the downwash from the main rotors (recirculation) can cause a significant change to the surrounding environment, it affects the performance greatly when the helicopter is close to the surface regardless of the altitude. All the test results were sent to MATLAB/Simulink through an UDP protocol. The aim for the flight test was to validate the design parameters from the other teams. The wind condition is assumed as negligible and zero payload was used unless specified. CG location is set at default throughout the test. The flight test has been divided into three parts, performance, stability[21], and handling quality[11]. This report will be focusing on performance only. All the procedure can be found from the respective flight test card in the Appendix.

4.1 Hovering Performance Test

Hover flight is an unique feature of the helicopter, and it is defined as a helicopter that is airborne at a given altitude over a fixed point on the ground regardless of the wind[22]. It allows the helicopter to perform a varied missions, such as plane guard and rescuing etc. Pilots consider hovering as the most challenging control, as helicopters are general dynamically unstable. Frequent control input and correction must be made by the pilot in order to maintain a constant altitude, heading, zero airspeed. The aim of this test is to determine whether the helicopter can reach a maximum capture altitude of 20000 ft as suggested.

Result

From the use of free hovering method[22], the helicopter were tested with different payloads whilst the pilot aim to hover as high as possible. Once the service ceiling (vertical speed is less than 100 kt/min) is reached[23], the pilot manually reduces the payload in the simulation software, and hence continue to climb to the next ceiling point. In reality, extra power is required due to the vertical drag caused by the fuselage body. X-Plane only allows the user to input the estimated fuselage body drag coefficient based on the frontal area[4]. Estimation of the vertical drag can be achieved by the strip approach, where the drag coefficient is calculated based on induced velocity from the rotor[24]. Therefore, the fuselage effect was not accounted during the hover, and the actual maximum hovering ceiling is lower than the one in simulation. The test result are shown in Figure 6. As expected, an increase in payload decreases the maximum hovering ceiling linearly. This is due to the decrease in excess power available as air becomes thinner at higher altitude. For zero payload case, the hovering ceiling is estimated as 24520 ft from the linear interpolation, and this value satisfies the mission profile. It is to be noted that the realistic hovering ceiling will be slightly lower due to the limitation of the X-Plane simulation. When the data is compared with the WRR02 team, the predicted absolute ceiling altitude is at 12000ft[25] during the full payload and it is over-predicted. This data cannot be verified as the engine output never reach the maximum available power and it only produces around 1000hp throughout the test. The experimental result showed that the helicopter can reach the capture altitude without payload.

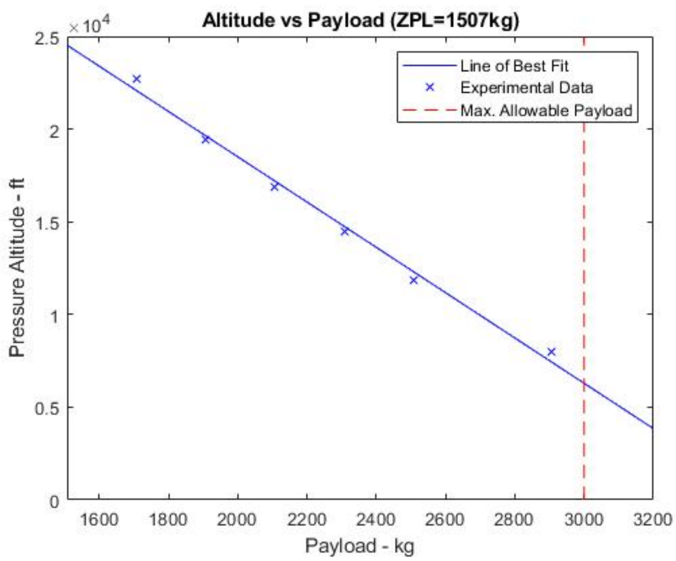


Figure 6: Prediction of maximum ceiling

4.2 Vertical Climb Performance Test

Vertical Climb is another special performance of the helicopter, which allows a high flexibility in different missions due to the present of excess power. A poor vertical climb performance can lead to the re-design of the model or alternation of the mission. A vertical rate of climb at 300 ft/min is considered as a minimum requirement[24]. The climb capability decreases as the altitude goes up as the power available decreases. Pilots are required to maintain a zero forward speed and a constant heading in order to obtain the relevant data from vertical climb test. The aim of this test is to determine whether the helicopter can meet the rate of climb as suggested from the mission profile, as well as the power available for climbing at different altitudes.

Result

From the use of continuous climb test method[22], the helicopter were able to be tested with different vertical climbing speed, and the rotor power were recorded as it passes a certain altitude.

As previously mentioned, the X-Plane does not include the fuselage effect during hovering or vertical climbing. Also, the fuselage drag was calculated based on forward velocity, therefore, the vertical drag is negligible during the hovering and vertical climbing. The vertical drag on the fuselage is typical 5 % of the weight, but it is estimated to be much higher due to its large rotor radius situated in the downwash field below the rotor[26]. Therefore, the experiment over-predicts the vertical climb performance. The test result are shown in Figure 7 and Table 4. Due to fuselage effect, the y-intercept should be lower in reality as extra increment of the power are required for the vertical drag.

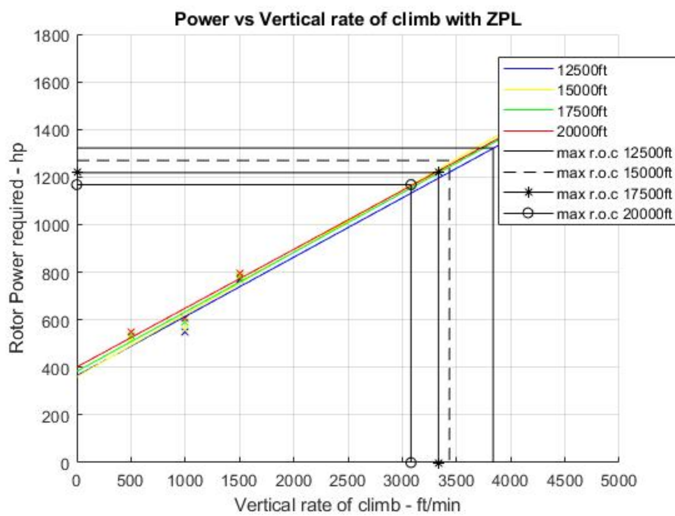


Figure 7: Power vs Vertical Speed

Altitude (ft)	Max. vertical rate of climb (ft/min)
12500	3700
15000	3400
17500	3300
20000	3100

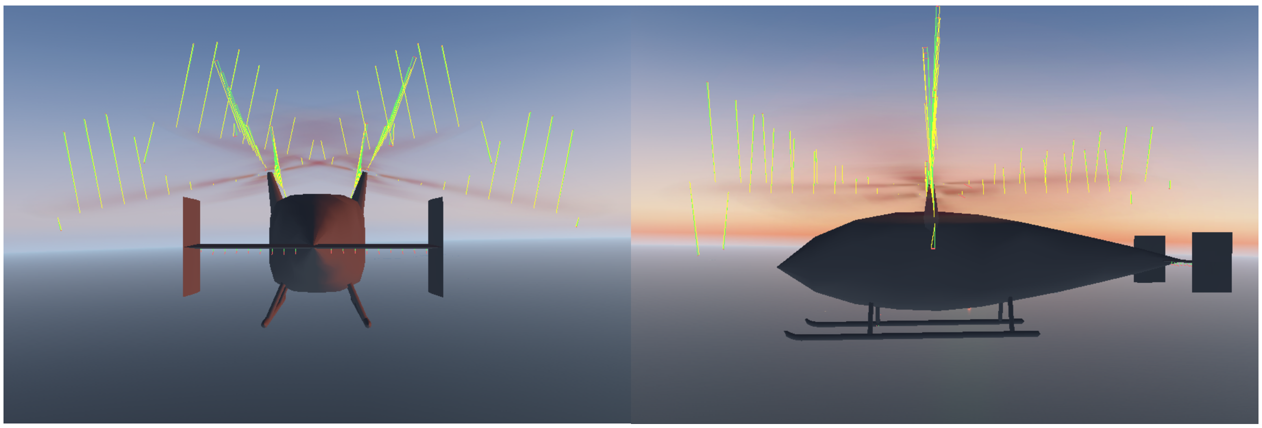
Table 4: Summary of the vertical climb test

4.3 Forward Speed Test

The forward speed test is simply the helicopter version of the performance test. Majority of the mission is travelling in a long distance and a high altitude, and therefore it is desirable to determine the least time, most efficient method for travelling. Poor performance will lead to a significant increase in fuel, cost and manpower, and would eventually affect the handling quality of the pilot. Pilots are required to maintain a constant altitude and essential to head at different airspeed and altitude in order to obtain data. The aim for this test is to determine the airspeed for max. rate of climb, max. range and level flight speed at different altitude. The max. L/D can be interpreted from these result.

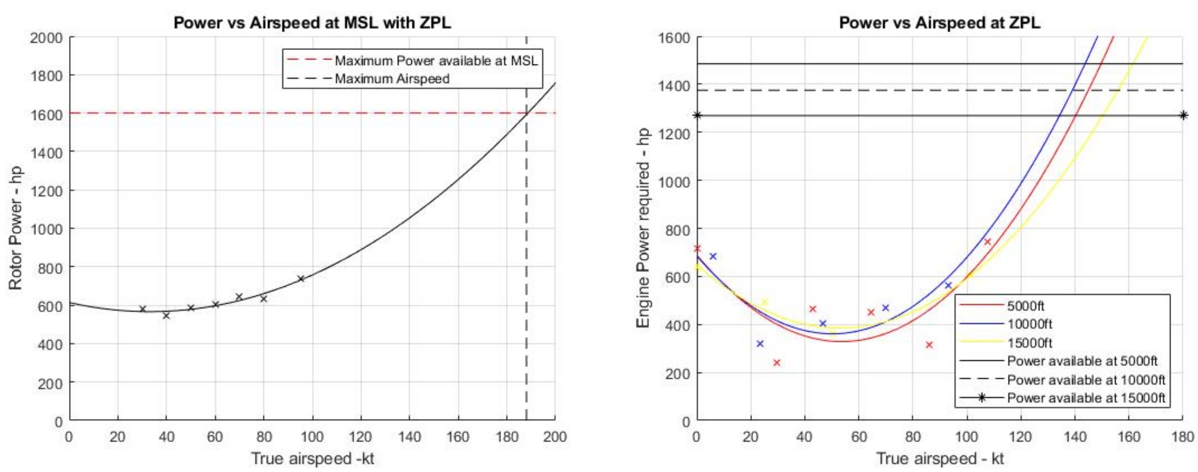
Result

The helicopter is tested at different altitudes and airspeeds, while the pilot records the respective power at designated point. Since the helicopter is designed as two overlapping intermeshing rotors, the aerodynamic symmetry were achieved at different forward flight. However, the intermeshing area affects the performance of the rotor. Researcher suggested that there is thrust reduction in the overlapping region due to the increased induced velocity[27][28]. This effect has not been simulated due to its oversimplified mathematical model in the X-Plane and it only calculates the individual force for each rotor blades as shown in Figure 11a. Therefore, the actual power for the forward flight is less than that of the experimental result. As expected from the theory, the parasite drag increases at a high speed and hence more power is required. The power curve at different altitudes is shown as a function of true airspeed, as shown in Figure 9. Although there is no direct relationship between the airspeed and the maximum rate of climb and range, it is possible to obtain these parameter graphically from the power curve. In addition, L/D ratio curve can be plotted. These plots can be used to compare its forward flight efficiency with different rotor or wing. The ratio can be simplified as, $\frac{L}{D} \approx \frac{WV_\infty}{P_{rotor}}$[24]. the summary is shown in Table 5. Compare with the theoretical data, WRR02 suggested that the airspeed for the maximum rate of climb to be around 58 knots for high altitude, and it shows a consistence in the model. In conclusion, this model meets the requirement of the mission profile as it flies between 185-225 km/h, Figure 10 indicates that the maximum lift-to-drag ratio is very low, in comparison with the typical modern helicopter which is around 6.



(a) Rear view of the model **(b)** Side view of the model

Figure 8: Force element break down in forward speed



(a) Mean Sea Level **(b)** Different Altitudes

Figure 9: Prediction of rotor power in forward flight

Altitude (ft)	5000	10000	15000
Airspeed for Max. r.o.c (kt)	55	51	53
Max. Level Flight Speed (kt)	150	140	150
Max L/D	1.9826	1.7407	1.8054
Airspeed at Max. L/D (kt)	74	72	83

Table 5: Summary of the forward performance test

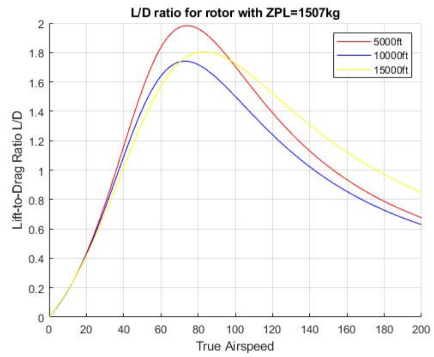


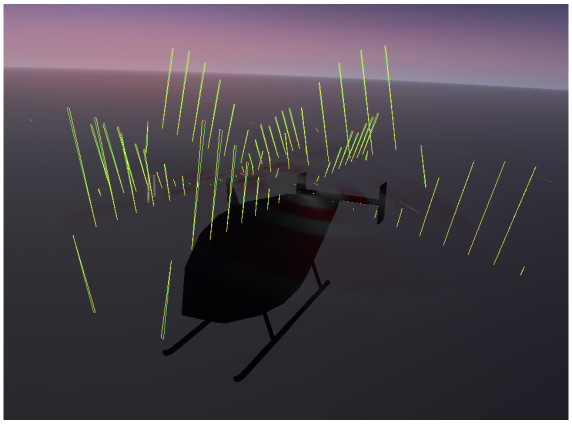
Figure 10: L/D vs Airspeed

4.4 Autorotation Forward Speed Test

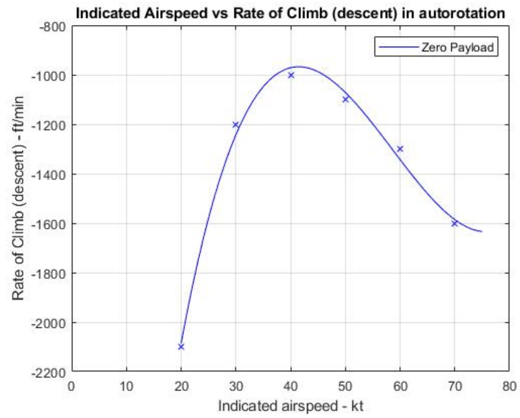
In the case of emergency, the pilot needs to identify a suitable landing zone away from the public. For such maneuver, the pilot can make use of autorotation for safe landing in the event of engine failure. Autorotation is defined as a self-sustained rotation of the rotor without the application of any shaft torque from the engine, the power to drive the rotor comes from the airstream upwards through the rotor during descend. Autorotation allows the rotor to produce thrust approximately equal to the helicopter's weight, hence a stable descent can be made to minimize the impact on the helicopter. To maximum the time in the air, the minimum rate of descent in autorotation needs to be identified. This is particularly important for this mission as it mainly operates away from land, hence increasing the flying time allow the vehicle to return to the nearest landing area in case of rotor failure. The aim for this test is to determine the autorotation rate of descent by switching off the engine from the altitude. The Height-Velocity curve (Deadman's Curve) has not been determined due to the inexperience and unprofessional judgement of test pilot[29].

Result

The test is carried out at 20000 ft without any payload, whilst the pilot switches off the engine to simulate the engine failure environment. As previously mentioned, the fuselage effect are not considered as lifting body or vertical drag body. Therefore, the actual rate of descent is slower in comparison with the experimental data from the flight simulator. The force vector during autorotation stage is shown in Figure 11a. The minimum rate of descent gives the pilot the maximum time to diagnose and attempt the correct the engine problem, or locating to the nearest landing site. Figure 11b shows the result from the test. The minimum rate of descent is 968 ms^{-1} and its respectively speed is 41 kts. These values were considered as low in comparison with the typical aircraft. This result does not show consistent with the theoretical model ,this is possible due to empirical method used[30] and the lack of experience in handling engine failure procure[31].



(a) Force vector during autorotation



(b) Estimation of rate of descent in autorotation

Figure 11: Autorotation performance test

5 Conclusion

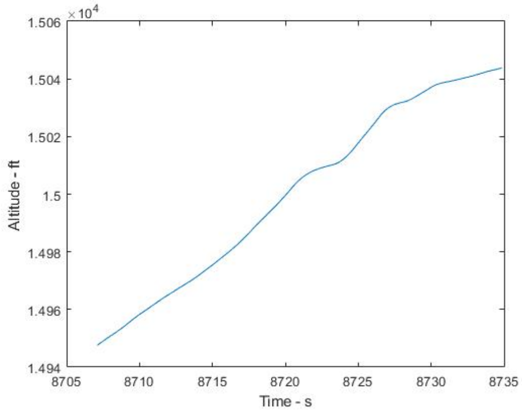
5.1 Aircraft Modelling

As previously mentioned, the X-Plane solely relies on the Blade Element Theory to generate a lift from the wing component. Precise geometry and aerodynamics characteristics were required in order to provide a realistic simulation. Due to the limitation of the airfoil maker, the airfoils were created only based on a few key parameters, and this does not produce a reasonable model. It was also observed that the supplied airfoil data in the plane maker is oversimplified. To obtain a better airfoil data, the X-Plane forum has suggested to use an external program ”FoilTrans” as created by the community members in 2009. It is a useful tool to convert the polar curve from XFOIL, XFLR5 and JavaFoil into an airfoil .afl file directly[32]. This method was not tested due to the incompatibility version of the software in 2020. Multiple sources should be used to cross-check the data in order to obtain a precise geometry of the airfoil. Due to the limited element available in the plane-maker, the BERP blade design cannot be modelled at the tip section. Although it is possible to create the twisting distribution of the blade, it can hardly resemble the performance of the BERP blade due to complex geometry at the tip section. It is possible to create the aerodynamic equivalence blade if time permitted.

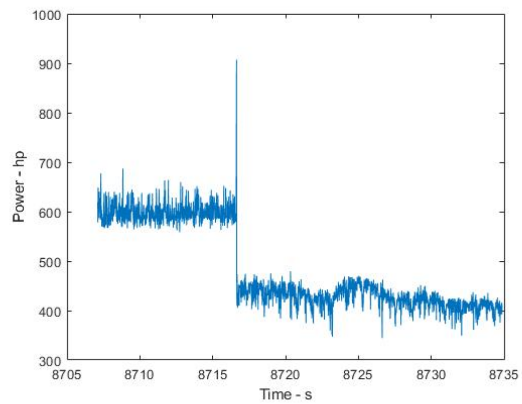
5.2 Flight Test

Due to the time constraint of the project, most performance tests were designed to match the requirement of the mission profile. In term of the performance test, the helicopter is tested with out-of-ground effect only due to the nature of the mission (e.g. mission occurs in air most of the time and without payload). In-ground test should be conducted in order to fully explore the performance of the helicopter as it is particularly important during takeoff and landing. Although all planned tests has been carried out, the data is not satisfactory as the performance of the engine was unstable. Investigation in the rotor engine can be conducted as a future work. Due to the limitation in MATLAB poly function, the curve cannot be simulated perfectly especially at low speed cruising. Other factor such as fuselage effect, intermeshing blade effect were not simulated in the X-Plane, therefore correction factor must be made in order to increase the accuracy of the model. Engineering Design Handbook from US Army[22] has suggested a detailed helicopter flight test including correction factor; accurate data can be made if there is a complete data of this model.

Finally, HeliDyn+ simulation tool[33] should be explored as an alternative simulation software if this model is to be re-designed in the future. It is specifically designed for the rotorcraft, and feature on the interaction between multiple rotors in which the X-Plane cannot provide.



(a) Altitude



(b) Power

Figure 12: High disturbance in engine power during vertical climb and hovering test

6 Future Improvement

To recap the role of WRR06, the team produces a realistic model in the flight simulator, and provides feedback on the performance and handling qualities to the other team. The goal is to identify the disadvantage or limitation of the flight simulator and comparing the model with the theoretical calculation. However, the result from simulation showed a poor agreement with the theoretical calculation due to limitation of the mathematical model in the X-Plane.

Although the final model closely resembled the geometry of the model, it did not fully reflect the performance of the helicopter. In particular, the tip section of the blade and the engine due to its detailed geometry and a number of empirical data. These two components dominated the performance of the model and it is crucial to recreate them explicitly. Lastly, alternative flight simulation should be used for rotorcraft design as the X-Plane does not provide a comprehensive result, especially in the intermeshing blade and fuselage.

To conclude, the helicopter is a very complex machine which requires a long time for investigation. To truly understand and optimize the behaviour of this model, a full repetitive flight test and iteration process should be conducted as future work.

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A X-Plane Model

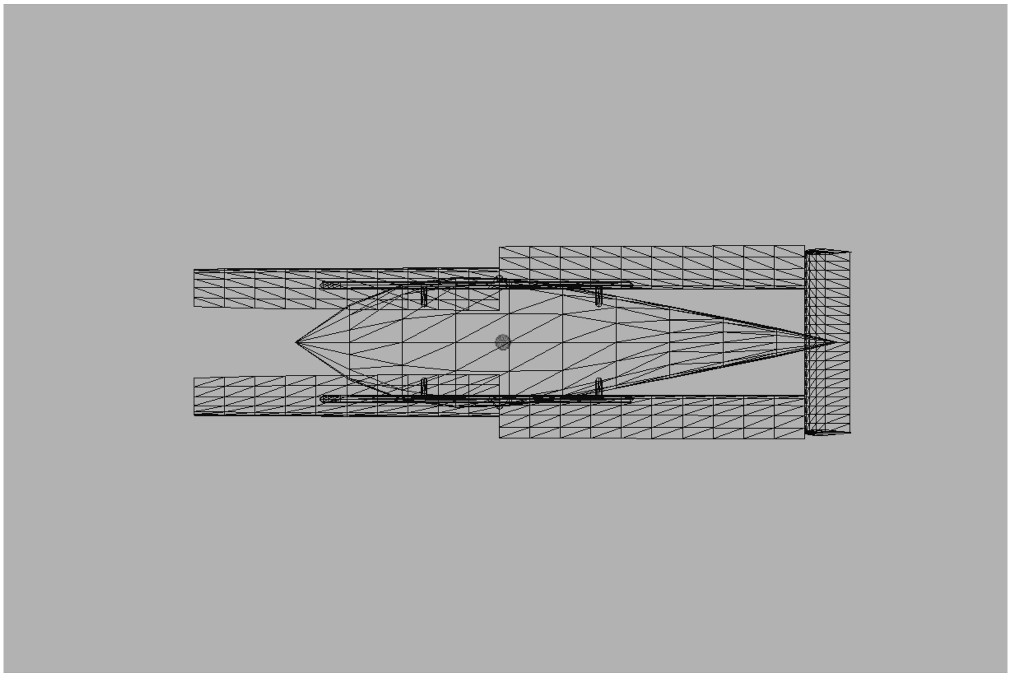

Figure 13: Top View of the model

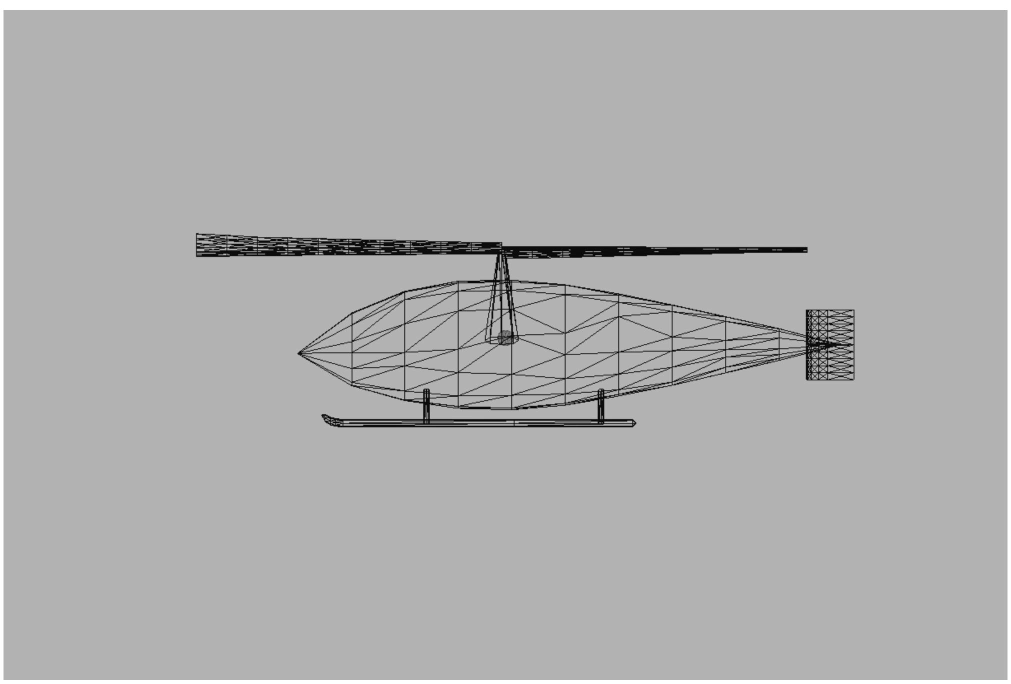

Figure 14: Side View of the model

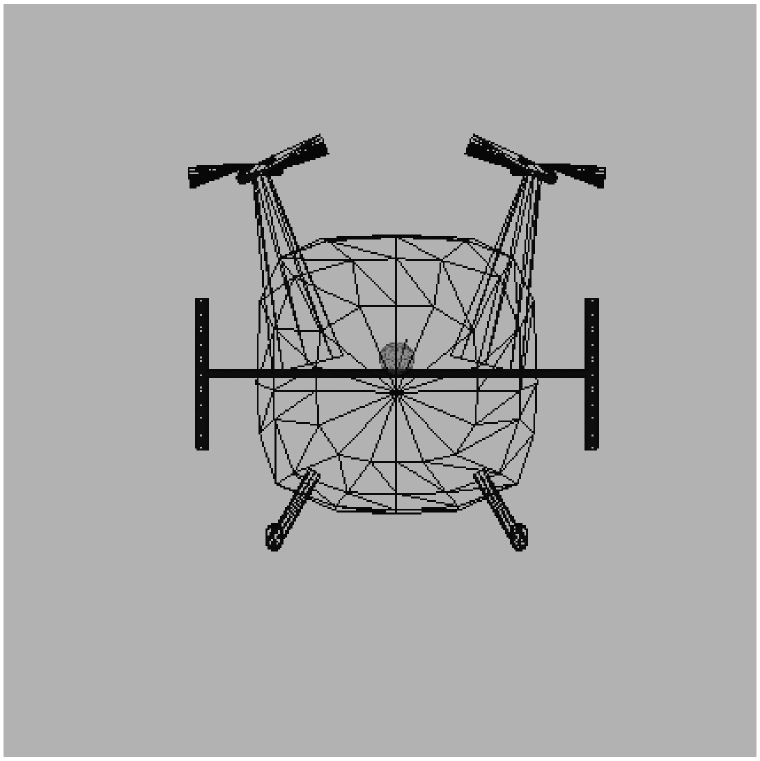

Figure 15: Front View of the model

B Aerofoil Design

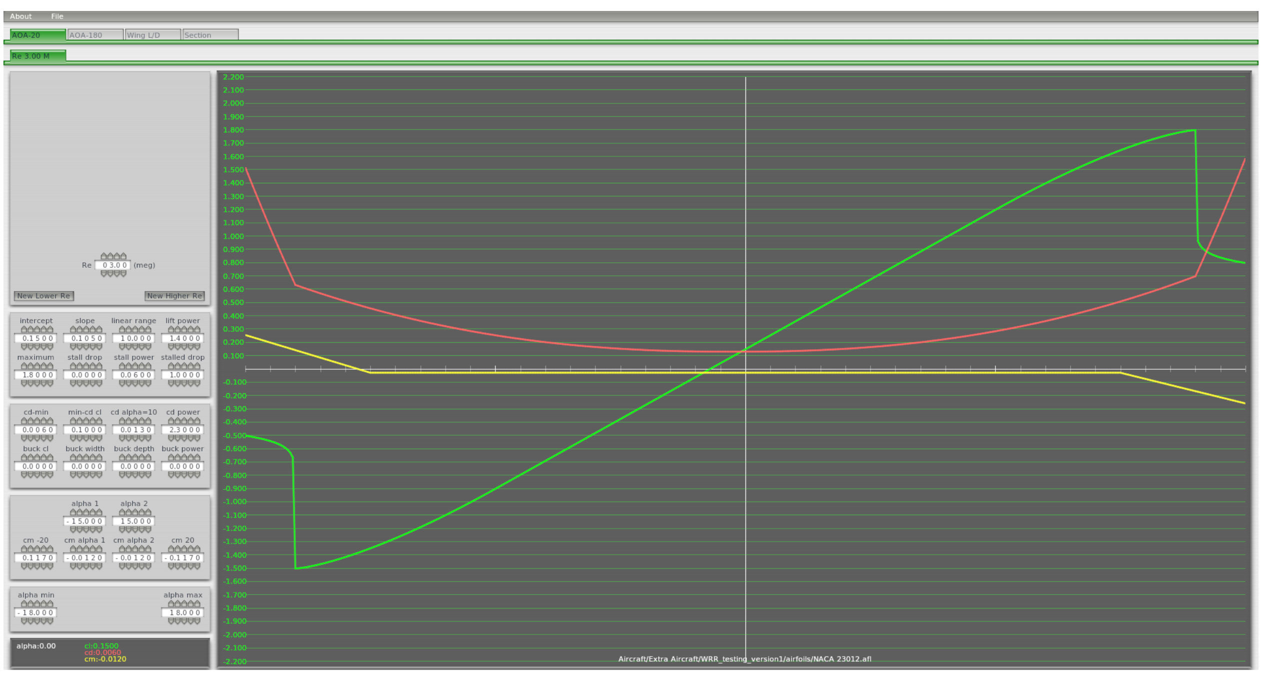

Figure 16: NACA 23012 (root of aerofoil in propeller) supplied by X-Plane

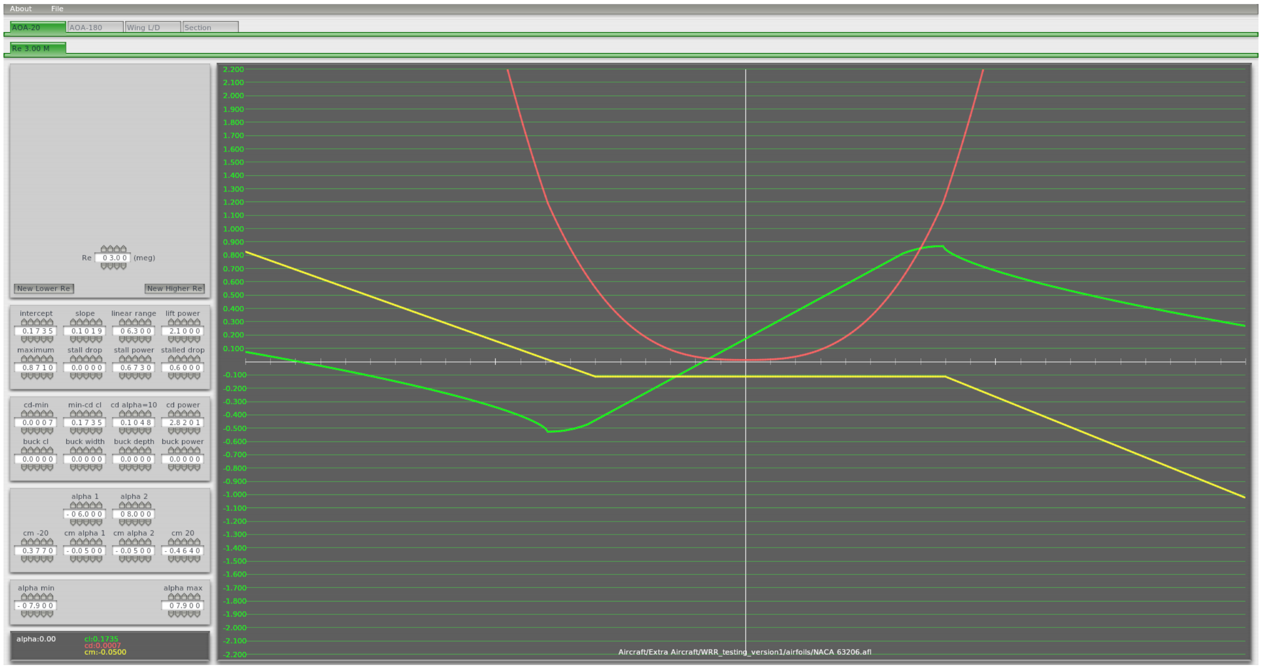

Figure 17: NACA 63206 (tip of aerofoil in propeller)

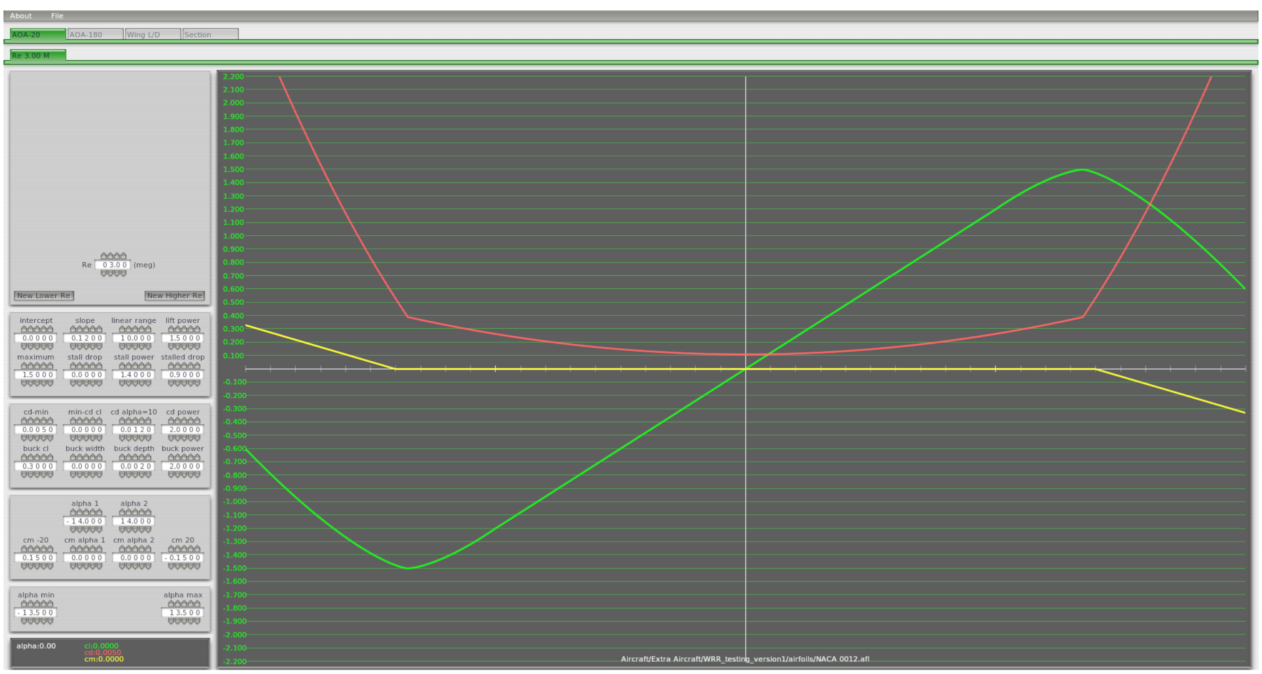

Figure 18: NACA 0012 (horizontal stabilizer) supplied by X-Plane

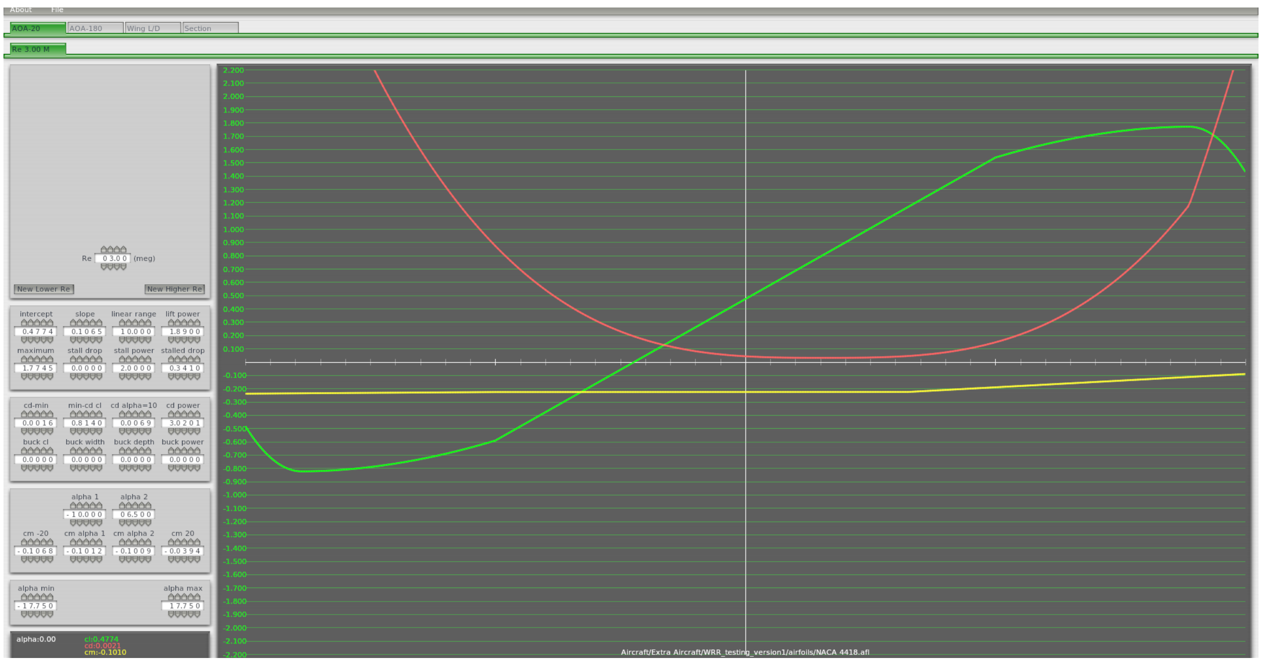

Figure 19: NACA 4418 (vertical stabilizer)

C Flight Test Card

<table>
<tr><th>Card No.</th><th colspan="2"></th><th colspan="2">Hovering Test</th></tr>
<tr><td>Pilot</td><td></td><td>Date & Time</td><td colspan="2"></td></tr>
<tr><td colspan="5">Objectives</td></tr>
<tr><td colspan="5">Find out the service ceiling for different payloads</td></tr>
<tr><td colspan="5">Configurations</td></tr>
<tr><td colspan="5">Weight: Varies (Payload + 1507 kg)
CG: Default</td></tr>
<tr><td colspan="4">Procedure</td><td>Check</td></tr>
<tr><td colspan="4">Stabilize the helicopter in static motion for 30 seconds at 2500 ft (within ±100ft)</td><td></td></tr>
<tr><td colspan="4">Record the power and weight</td><td></td></tr>
<tr><td colspan="4">Increase the altitude until the vertical speed is less than 100 kt/min</td><td></td></tr>
<tr><td colspan="4">Record down the</td><td></td></tr>
<tr><td colspan="4">Plot Power vs Altitude graph and label the maximum gross weight</td><td></td></tr>
<tr><td colspan="5">Data Recording</td></tr>
<tr><td>Gross Weight (kg)</td><td>RPM</td><td>Torque (%)</td><td>Power</td><td>Service Ceiling</td></tr>
<tr><td>200</td><td></td><td></td><td></td><td></td></tr>
<tr><td>400</td><td></td><td></td><td></td><td></td></tr>
<tr><td>600</td><td></td><td></td><td></td><td></td></tr>
<tr><td>800</td><td></td><td></td><td></td><td></td></tr>
<tr><td>1000</td><td></td><td></td><td></td><td></td></tr>
<tr><td>1400</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="5">Notes:</td></tr>
<tr><td colspan="5"></td></tr>
</table>

Figure 20: Hovering Test

Card No.			Vertical Climb Test	
Pilot		Date & Time		
Objectives				
Find out the power required to maintain constant rate of climb at different altitude				
Configurations				
Weight: 1507 kg CG: Default				
Procedure				Check
Stabilize the helicopter in static motion for 30 seconds at 12500 ft (within ±100ft)				
Increase the power whilst maintain zero airspeed and constant desired vertical climb speed				
Record the RPM/Rotor Speed and Torque between ±500 ft of desired altitude until it reached 20000 ft				
Repeat the task with 5000ft increment until 25000ft				
Plot Power vs Vertical Climb Speed at different altitudes and label the maximum power available line				
Data Recording				
Altitude (ft)	Vertical Climb Speed (ft/min)	RPM	Torque (%)	Power
12500	500			
15000				
17500				
20000				
12500	1000			
15000				
17500				
20000				
12500	1500			
15000				
17500				
20000				
12500	2000			
15000				
17500				
20000				
Notes:				

Figure 21: Vertical Climb Test

Card No.			Forward Speed Test		
Pilot		Date & Time			
Objectives					
Find out the airspeed for max. rate of climb , range and max. L/D at different altitudes					
Configurations					
Weight: 1507 kg CG: Default					
Procedure					Check
Stabilize the helicopter at 5000 ft and allow hovering for at least 30 seconds					
Record the RPM and Torque at zero IAS					
Increase forward speed to 40 kt whilst maintain constant altitude (±100 ft)					
Repeat the task with 20 kt forward speed increment until 100 kt					
Repeat the task with 5000 ft increment until 25000 ft					
Plot Power vs True Airspeed and label max. r.o.c. , range and level flight speed					
Plot L/D vs airspeed at different altitude and locate the maximum L/D					
Data Recording					
Altitude (ft)	Forward Speed (kt)	RPM	Torque (%)	Power	L/D
5000	0				
	20				
	40				
	60				
	80				
10000	0				
	20				
	40				
	60				
	80				
15000	0				
	20				
	40				
	60				
	80				
20000	0				
	20				
	40				
	60				
	80				
Notes:					

Figure 22: Forward Speed Test

Card No.		Autorotation Test	
Pilot		Date & Time	

Objectives
Find out the minimum autorotation rate of descent
Configurations
Weight: 1507 kg CG: Default

Procedure	Check
Stabilize the helicopter at 20000 ft and allow hovering for at least 30 seconds	
Switch on the engine failure mode to simulate zero power condition	
Lower the collectives completely to perform autorotation and pitch up	
Maintain constant heading at 20 kt IAS (5 kt error)	
Between 1000 ft and 19000 ft, record the time taken to descent every 1000 ft	
Repeat the task with 20 kt increment until the speed reaches 120 kt	
Calculate the mean rate of descent for each airspeed	
Plot rate of climb (descent) against airspeed	

Data Recording

Airspeed (kt)	Time Taken to descend 1000 ft (s)					Mean rate of descent (ft per second)
20						
40						
60						
80						
100						
120						

Notes:

Figure 23: Autorotation Test